MILITARY SPECIFICATION

TRUCK TRACTOR: COMMERCIAL 85,000 POUNDS GVWR, 8X6, XM911

This specification is approved by Army Tank-Automotive Materiel Readiness Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a standard commercial, diesel engine driven, eight wheel, six-wheel drive, truck tractor having a minimum gross vehicle weight rating (GVWR) of 85,000 pounds for on/off road operating conditions and which has been commercially proven. The term "standard commercial truck tractor" is defined in Section 6 of this specification. Vehicles under this specification shall be procured only from a manufacturer who meets the standard commercial truck tractor and commercially proven components criteria. The vehicle will be warranted by the manufacturer as specified in the procurement documents.

1.2 General use. This vehicle will be primarily used as the prime mover for the US Army M747 Semitrailer to a Gross Combination Weight (GCW) as specified herein. (See Appendix A and B for M747 technical characteristics). The truck must be capable of operating in on/off road conditions and must be capable of withstanding the strains, shocks, vibrations, and other detrimental conditions incident to off road operation with minimum loss of time for maintenance and repair. All parts of the truck shall be of such size, material and strength as to properly sustain the maximum allowable GVWR and GCWR stated herein with a maximum working efficiency and minimum wear during operation.

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

RR-W-1817	-	Warning Device, Highway, Triangular, Reflective.

Military

MIL-E-52835	-	Enamel, Modified Alkyd, Camouflage, Lusterless.
MIL-E-52798	-	Enamel, Alkyd, Camouflage.

STANDARDS

Military

MIL-STD-1223	-	Administrative Wheeled Vehicles Treatment, Painting, Rustproofing, Undercoating, Identification Marking, Data Plates and Warranty Notice Standards.
MS-35000	-	Batteries.
MS-51118	-	Pintle Assembly, Towing: 40,000 lbs. Capacity, Manual Release.
MS-53064	-	Tank, Combat, Full Tracked: 105 MM Gun, M60A1.
MS-75020	-	Connector, Plug, Electrical - 12 Contact, Intervehicular, 28 Volt, Waterproof.
MS-75021	-	Connector, Receptacle, Electrical - 12 Contact, Intervehicular, 28 Volt, Waterproof.
MS-500048	-	Towbar, Motor Vehicle: Light, Medium, and Heavy Duty (For Vehicles 2 1/2 to 50 Tons and Heavier.)

DRAWINGS

Army

7728811	-	Harness Assy Intervehicular.
7731428	-	Cover Assembly, Receptacle.
11677011	-	First Aid Kit.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuringg activity or as directed by the Contracting Officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

DEPARTMENT OF TRANSPORATION

Federal Motor Vehicle Safety Standards.
Federal Motor Carrier Safety Regulations.

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(Application for copies should be addressed to the Department of Transportation, Federal Highway Administration, Washington, D.C. 20591.)

ENVIRONMENTAL PROTECTION AGENCY
Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.
Compliance with Interstate Motor Carrier Noise Emission Standards.

(Application for copies should reference "Code of Federal Regulations 40CFR and the Federal Register", and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington D.C. 20402.)

OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION
Noise Emission Standards for Transportation Equipment Part III.

(Application for copies should reference Federal Register, Volume 40CFR, Part 205, Title 40, Chapter I, and should be addressed to US Department of Labor, Occupational Safety and Health Administration, Washington D.C. 20210.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)
SAE Standards and Recommended Practices.

J551	-	Measurement of Electromagnetic Radiation from Motor Vehicles (20-1000 MH_z).
J552	-	External Electromagnetic Radiation Suppressors.
J598	-	Sealed Lighting Units for Construction and Industrial Machinery.
J645	-	Transmission Definitions.
J682	-	Rear Wheel Splash and Stone Throw Protection.
J688	-	Truck Ability Prediction Procedure.
J816	-	Engine Test Code - Spark Ignition and Diesel.
J848	-	Fifth Wheel Kingpin - Heavy Duty.

(Information as to the availability of copies may be obtained from The Society of Automotive Engineers, Inc., Warrendale, Pa. 15086.)

THE TIRE AND RIM ASSOCIATION, INC.
Yearbook

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 W. Market Street, Akron, Ohio 44313.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Standard vehicle components, parts and accessories. The standard commercial truck tractor, hereinafter referred to as "the vehicle", and its components, parts and accessories shall be standard or optional commercial items which meet or exceed the requirements of this specification. The vehicle shall be complete in all respects as normally delivered to the commercial heavy truck transport industry. All chassis components shall be as represented in the chassis manufacturer's current technical data and all body and mounted equipment shall be as represented in the body and equipment manufacturer's technical data. The technical data herein referred to shall be identical to the specifications and technical material furnished to the authorized company representatives for use in selection of vehicle models and components and shall be on file in the engineering offices of the procuring activity. The vehicle and all of its components shall have currently published rated capacities consistent with both the on/off road usage. All power train components shall be certified by the component manufacturing companies as being compatible with and properly matched with all related or affected components assembled to meet the specifications stated herein.

3.1.1 Special requirements. In addition to standard commercial vehicle and components specified in 3.1 above, the vehicle shall be furnished with the special items as specified in 3.1.1.1 through 3.1.1.5 below and Para 3.2.9.

3.1.1.1 Painting, marking and data plates.

a. Painting. All metal or plastic surfaces not normally painted shall be treated to provide a limited reflectance surface. The final vehicle exterior paint will be of best paint quality having color, gloss and spectral reflectance properties of lusterless forest green as described in MIL-E-52835 and MIL-E-52798. Final exterior paint shall be applied to all exposed surfaces visible from above or the horizontal at a distance of 20 feet.

b. Instruction, caution, identification, operating and data plates. Instruction, caution, identification, operating and data plates shall be provided and installed in accordance with the manufacturer's normal commercial practice. Military model number, nomenclature, national stock number (NSN), contract number, date of manufacture, manufacturer's serial number and US Army Registration number shall be imbedded or embossed on an additional metal identification plate and installed in the vehicle cab in a readily visible location. External commercial markings shall be lusterless black.

c. Exterior model name plates and exterior trim. Exterior model name plates may be eliminated, providing no holes are left.

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3.1.1.2 Rustproofing. Rustproofing shall be furnished. Rustproofing shall be in accordance with or equal to MIL-STD-1223.

3.1.1.3 Towing and tiedown provisions. Vehicle shall be equipped with heavy-duty towing, and tiedown devices which shall be adequate for highway, rail, sea and air transport.

3.1.1.3.1 Towing eyes. Two towing eyes shall be furnished and installed on the front of the vehicle and two towing eyes shall be furnished and installed on the rear of the vehicle. Each towing eye and its mounting shall withstand a load of at least 85,000 pounds without failure or permanent deformation when the load is applied at an angle up to 45° from the longitudinal axis. The towing eyes shall be of a size such that the vehicle can be towed with the medium duty towbar described in MS-500048.

3.1.1.3.2 Pintle. A towing pintle at the rear of the vehicle shall be furnished. The pintle assembly shall be of the swivel type and conform to the size and strength described in MS-51118. The assembly shall be furnished with mounting flanges and lubrication fitting. The pintle assembly mounting surface shall be forward but not more than 4 inches forward of the rear most part of the vehicle. The mounting of the pintle assembly shall include reinforcements to transfer pintle loads directly to the web of the chassis frame.

3.1.1.3.3 Tiedowns. The vehicle shall be furnished with 2 tiedowns on each side of the vehicle and be capable of withstanding a load of at least 42,000 lbs. without failure or permanent deformation when the load is applied at an angle up to 45° from the longitudinal axis. The tiedown devices plus the towing eyes defined in para 3.1.1.3.1 shall permit tiedown of the vehicle to the floor (or deck) of the transportation medium in such a manner as to prevent shifting or movement in any direction.

3.1.1.4 Kits. The vehicle shall operate in accordance with the specification requirements and the manufacturer's technical data after field installation of and during the use of the kits specified herein. When specified (see 6.2), the manufacturer shall be required to furnish an arctic kit. When applied, the kit should permit the vehicle to start after 24 hour cold soak and operate, without abnormal wear under the extreme climatic conditions specified in paragraph 3.1.1.5.1. The kit must be of normal commercial application and use. The proper application and use of the kit will in no way affect the warranty or life expectancy of the vehicle or its components. If external electrical power is required for operation of elements of the kit, it shall not exceed 110 volts AC.

3.1.1.5 Climatic conditions. The vehicle shall be capable of operation in ambient temperatures from -25°F to /125°F without the use of kits.

3.1.1.5.1 Climatic conditions, extreme. When specified (see 6.2), the vehicle shall be capable of operation in ambient temperatures of -25°F to -50°F with manufacturer's arctic kit (see 3.1.1.4) installed.

3.2 General design.

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3.2.1 Federal Motor Vehicle Safety Standards. Vehicles and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture, except as specified herein.

3.2.2 Air/Sound Pollution.

a. Vehicles shall comply with the Environmental Protection Agency regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of invitation for bid.

b. The interior sound level shall conform to Federal Motor Carrier Safety Regulation 393.94. The vehicle exterior sound level shall conform to the Interstate Motor Carrier Noise Emission Standards of the Environmental Protection Agency when tested in accordance with the regulations of the Department of Transportation, Part 325.

c. The exterior noise level shall not exceed the noise emission standards for Transportation equipment. Part III, Chapter I, Part 205, Title 40.

3.2.3 Curb weight. Curb weight shall include weight of chassis and cab, with all attachments, accessories and equipment, fifth wheel, full complement of fuel, winches, lubricants, coolant and driver.

3.2.4 Gross vehicle weight. Gross vehicle weight (GVW) is defined as the curb weight plus the load imposed upon the fifth wheel. The vehicle shall have a 5th wheel load of 46,000 lbs.

3.2.5 Gross vehicle weight rating. Gross vehicle weight rating (GVWR) is defined as the sum of the curb weight and the maximum 5th wheel load that can be imposed without exceeding the rated capacity of any component in the truck tractor for both on and off road operating conditions. The GVWR shall be no less than 85,000 pounds.

3.2.6 Gross combination weight. Gross combination weight (GCW) is defined as the truck tractor curb weight, and weight of a semitrailer with payload.

3.2.7 Gross combination weight rating. Gross combination weight rating (GCWR) is the sum of the GVWR and the maximum semitrailer towed load (weight on semitrailer axles) that can be imposed without exceeding the rated capacity of any components in the truck tractor for both on and off road operating conditions. The GCWR shall be no less than 190,000 lbs.

3.2.8 Ratings. Vehicle ratings shall be manufacturer's published rating for on/off road operating conditions. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, component manufacturer's verification of ratings must be submitted to the engineering office of the procuring activity.

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3.2.9 Dimensions. Dimensions of the vehicle in the unloaded condition shall be as follows:

a. Fifth wheel height - 64 inches plus 1 inch, minus 0 inches.

b. Overall height - shall not exceed 144 inches.

c. Overall width - no more than 115 inches.

d. Swing clearance (from centerline of kingpin to rear tires and chassis frame rails) - 103 inches maximum.

e. Swing radius (from centerline of kingpin to rear point of obstruction (semitrailer nose to tractor components) forward of the 5th wheel - 37 inches minimum. There shall be no obstruction rear of a transverse line 37 inches forward of the kingpin.

3.2.10 Fuel and lubrication. Vehicles shall be operable with applicable standard military fuel and lubricants without adverse effect on vehicle components or warranty.

3.3 Performance. Requirements shall be met with truck tractor loaded to specified GCWR. Minimum gradeability and speed requirements are as follows:

3.3.1 Speeds and gradeability.

a. Shall start on and climb a 20% grade (adequate traction assumed). If a torque converter is provided the vehicle must start and climb a 20% grade without exceeding the multiplication and converter efficiency factor recommended by the manufacturer for continuous duty. If a torque converter is not provided the vehicle must be capable of starting on a 20% grade without deviating from normal starting procedures.

b. Shall maintain a speed of 14 MPH on a 3% grade (if a two speed transfer case is provided, this performance must be in high range).

c. Shall be capable of maintaining a speed of 43 MPH on level roads. Vehicle shall meet above requirements based upon calculations in accordance with SAE J688 using the factors contained in paragraph 6.4, Truck Ability Prediction Procedures using Class 1 roads and actual performance demonstrations pulling the M747 Semitrailer with a 60 ton payload. Performance will be demonstrated at a Government site.

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3.3.2 Maximum geared speed. Maximum geared speed at engine full load governed speed shall be not less than 43 MPH. Engine governed speed will not exceed the maximum RPM rating specified by the engine manufacturer.

3.3.3 Grade and slope operation. The vehicle shall function headed up and headed down slope on longitudinal grades of at least 20% and side slopes up to 10% with each side of the vehicle up slope. As a result of the operation, no evidence of faulty lubrication, leakage or other malfunction shall be found.

3.3.4 Service brakes. Service brakes shall meet the requirements of Section 393.52 of Federal Motor Carrier Safety Regulations.

3.3.5 Parking brake. A parking brake shall be furnished and shall conform to the Federal Motor Carrier Safety Regulation 393.41 and brake its own GVWR on a 20% grade.

3.4 Chassis components.

3.4.1 Engine. Engine furnished shall be the chassis manufacturer's standard or optional engine for the commercial model truck which meets or exceeds the requirements of this specification. Vehicle shall be equipped with a liquid cooled compression ignition, two-stroke or four-stroke cycle diesel engine, with not less than six cylinders. Engine net horsepower figures used in performance prediction calculations shall be determined in accordance with SAE J816. Engine shall be rated not less than 400 gross horsepower intermittent automotive rating.

3.4.1.1 Heavy duty coolant system. A heavy-duty cooling system shall be furnished. The cooling system shall maintain the top tank coolant at a temperature recommended by the engine manufacturer but not more than 220°F when operated under the conditions outlined herein in all speed ranges. The radiator pressure cap shall not exceed 10 PSI. The cooling system shall pass the engine manufacturer's cooling test in the installed configuration in ambient air temperature of 125°F. If the radiator cooling system is used to cool the retarder and/or transmission systems, the cooling system shall be capable of adequately cooling these components in addition to the engine as described above. It shall have thermostatic controls for controlling the flow of engine coolant through the radiator. Cooling system shall be adequate to also cool the torque converter when operating at 70% efficiency.

3.4.1.1.1 Fan clutch. A thermostatically controlled fan clutch shall be provided with a positive lock-up in case of failure. Vehicle performance specified elsewhere in this specification shall be with fan clutch engagement such that fan is operating.

3.4.1.1.2 Coolant system and indicators. The coolant system shall include a de-aeration system, or a surge tank, or a coolant recovery reservoir of not less than 5% total coolant system capacity. A high coolant temperature warning light and buzzer shall be provided. The reservoir shall be located in a position readily accessible for inspection and service.

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3.4.1.2 Oil filter. A full flow type oil filter as specified by the engine manufacturer and a bypass oil filter if recommended by the engine manufacturer shall be furnished for maximum engine protection.

3.4.1.3 Governor. Engine governor shall be furnished and set and sealed to limit engine to engine manufacturer's maximum recommended operating speed. Governor control linkage (foot throttle) shall be adjusted to permit the full range of engine-governed rpm. The accelerator control system shall conform to FMVSS 124.

3.4.1.4 Engine air induction system. The engine air induction system, as fabricated and properly installed from air cleaner to engine intake, and air compressor shall be free from all weld slag and foreign material. An air cleaner indicator shall be provided.

3.4.1.5 Engine starting. Engine starting equipment consisting of glow plugs in each cylinder or ether injection system shall be provided. If an ether system is furnished it shall be of the automatic injection, engine temperature limited type controlled from the driver's area and inoperable with the engine running.

3.4.2 Electrical system. Electrical system shall be in accordance with Motor Carrier Safety Regulations 393.27 through 393.31 and 393.33.

3.4.2.1 Starting and lighting system. A 24-volt starting and lighting system, with not less than a 50 ampere alternator, shall be furnished. Alternator shall provide sufficient amperage (when operating at engine idle speed) to operate all electrical components of the tractor and semitrailer.

3.4.2.2 Lighting. All vehicle lights, reflectors, and wiring shall be as specified herein. Lights and reflectors shall not be mounted on vehicle bumpers (FMVSS 108 headlight height does not apply). Rear lighting shall be mounted in a protective location to preclude damage when interfacing with a towed vehicle.

3.4.2.3 Turn signals. The vehicle shall be equipped with turn signals in accordance with FMVSS 108.

3.4.2.4 Lighting receptacles and cables.

3.4.2.4.1 Lighting receptacles, 24 volt. One lighting receptacle conforming to MS-75021-2 with cover assembly conforming to drawing 7731428 shall be furnished. The receptacle shall be mounted in an accessible location so that the intervehicular cable specified in 3.4.2.4.2 will be mounted as specified in 3.4.21.5. The receptacle shall be wired to provide electrical power to terminals B, D, E, J, K and L of MS-75021-2.

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3.4.2.4.2 Lighting cable, 24 volt. A 24-volt, 12 conductor lighting cable conforming to the cable requirements of drawing 7728811 shall be furnished. The lighting cable shall be at least 10 feet in length and both ends of the cable shall be equipped with connectors conforming to MS-75020-1 and MS-75020-2.

3.4.2.5 Batteries. A minimum of four 12-volt batteries shall be furnished. Batteries shall be in accordance with type 6TN of MS-35000.

3.4.2.6 Radio interference suppression. Vehicle shall be suppressed to limit electromagnetic radiation in accordance with SAE J551. Suppressors, where used, shall conform to SAE J552.

3.4.2.7 Horn. Manufacturer's standard twin trumpet 19" minimum air horns and standard city horn provided in normal commercial practice for this class of vehicle shall be furnished.

3.4.2.8 Flashing beacon. An amber flashing strobe vehicular warning light shall be provided. Light shall be cab mounted with reinforcement at point of attachment on vehicle centerline for 360 degree visibility. The beacon light shall be operated by means of a separate switch located on instrument panel and marked to indicate function of switch.

3.4.2.9 Floodlights. Two swivel-type sealed-beam floodlights shall be provided at the rear of the cab. Lights shall conform to SAE J598, Trade No. 4578 or 4593. Floodlights shall be not less than 4 inches in diameter and 1500 candlepower and be provided with an on/off switch marked to indicate the function of the switch in the vehicle cab accessible to the driver.

3.4.3 Fuel system. Fuel system shall conform to Federal Motor Carrier Safety Regulations 393.65 and 393.67.

3.4.3.1 Air cleaner. Manufacturer's heavy duty dry type air cleaner as recommended by the engine manufacturer adequate for off road operation with a safety element shall be furnished.

3.4.3.2 Fuel tanks. Truck tractor shall be equipped with fuel tank(s) adequate to provide a 300 mile operating range at rated GCW traveling at an average of 40 MPH, but of not less than 150 gallon capacity. When more than one tank is furnished means shall be provided to assure equalized fuel level in both tanks. A shut-off valve between the 2 tanks shall be furnished. Filler neck opening must be at least 3 inch diameter.

3.4.4 Exhaust system. The exhaust system shall conform to Motor Carrier Safety Regulation 393.83. The exhaust system as installed shall be gas tight and leak proof to prevent the accumulation of exhaust gas in the occupied areas in accordance with best commercial practice. The tail pipe(s) shall be vertically stacked at the rear of the cab. The tail pipe(s) shall be provided with

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a corrosion resistance rain cap to preclude entry of rain water when vehicle is not operating. Exhaust mufflers and tail pipes shall be furnished with adequate guards to prevent personnel contact while operating winches, entering and exiting cab and normal entry and exit of rear deck.

3.4.5 Gear train.

3.4.5.1 Transmission. The transmission shall be a hydrodynamically coupled automatic, semi-automatic or power assist as defined in SAE J645b with not less than five (5) forward gear ranges, and must be capable of upshifting at full throttle without reducing engine speed and without using a fly-wheel clutch. It must have a gear range capable of meeting the performance specification as stated herein. The transmission and transfer case (if multi-speed) must be capable of shifting or being shifted without the use of a clutch pedal.

3.4.5.2 Retarder. A power train retarding device shall be provided which develops at least 75% of the rated horsepower output of the engine. In addition, if the retarder has interface with the engine, it shall be approved by the engine manufacturer and shall not by its operation cause an exhaust back-pressure which exceeds the engine manufacturers allowable limit. Retarder will be of a variable self-cancelling type operated by a foot pedal to be operated by the driver. If such a foot pedal type is not available a self-cancelling foot control will be acceptable if provided with a red warning light which goes on when the retarder is engaged.

3.4.6 Transfer case. A transfer case is required to provide drive to the front axle. The transfer case shall be provided with lock-up for positive drive of all axles under off-road and hill climbing conditions. This lock-up control shall also lock-up the inter-axle differential (see 3.4.9.1) at the same time. If a 2 speed transfer case is provided, it must be capable of providing a minimum speed of 20 MPH at governed RPM in low range. The transfer case shifts may be accomplished with the vehicle stopped.

3.4.7 Frame. Chassis frame shall be manufacturer's heavy-duty standard for the type vehicle furnished. Manufacturer must warrant the frame and frame members for on/off road operation at maximum GCWR plus winching operations at maximum capacity but will not be less than 3,100,000 in. lbs. RBM per rail.

3.4.8 Suspension. Vehicle shall be equipped with a suspension system with components having a rated capacity at least equal to that specified in 3.2.5 and to the load imposed on each member, measured at the ground, with vehicle loaded to specified GVW. When spring capacity is rated at the spring pads, unsprung weight shall be deducted.

3.4.9 Axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, with vehicle loaded to specified GVWR (see 3.4.21.1).

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3.4.9.1 Rear bogie. The rear bogie on the vehicle shall be of the four-wheel type, complete with axles, torque rods, interaxle differential with lock-up (or inter axle power divider) and all other necessary parts. Axle gear ratios shall provide performance as specified in 3.3.1 and 3.3.2. This rear bogie shall be rated at not less than 65,000 pounds measured at the ground.

3.4.9.2 Front axle. The front axle shall be a driving axle with differential rated at not less than 20,000 lbs.

3.4.9.3 Pusher axle. The second axle on the vehicle shall be of the pusher type with single wheels and located ahead of the rear bogie and be of the non-driving type capable of being lifted off the ground 6 inches by the operation of an air control selector switch readily accessible to the driver in the cab. When the pusher axle is on the ground and vehicle fully loaded, the axle loads on the last 3 axles shall be as equal as practicable. The pusher axle and suspension shall have a capacity of at least 20,000 lbs. An air regulator control valve (with air pressure gage and loading chart) in the cab readily accessible to the driver shall be provided to adjust the pusher axle air suspension pressure for equalized loads. Pusher axle shall be equipped with brakes.

3.4.10 Wheels, rims, tires, and tubes. Vehicle shall be equipped with single front and dual rear wheels. Rims and tire ratings shall conform to Tire and Rim Association recommendations for the type and size of tires furnished. Wheels shall be of the disc type. All tires installed on the vehicle, and all rims and wheels used on the driving axle shall be identical. Rims shall be at least 24 x 10 inch size.

3.4.10.1 Tires. Tires shall be a minimum of 14.00 x 24, 18 ply with on/off road tread. The tire tread pattern shall have rib grooves for directional stability and emphasized non-directional shoulder lugs for off-road operation. Tires shall be of rated capacity (with tire pressure not greater than 95 psi) at least equal to the load imposed on each tire, measured at each wheel at the ground, with vehicle loaded to specified GVW. The tires must be of the continuous duty type meeting Table HTB-1B of the 1976 Tire and Rim Association Year Book.

3.4.10.2 Spare tire and wheel assembly. A spare wheel and tire assembly capable of being installed on either front or rear dual axles is to be provided with each vehicle. A spare tire carrier shall be furnished and mounted in readily accessible location when the tractor is fully loaded with M747 Semitrailer. The tire carrier shall be mounted in such a fashion that the tire and wheel can be removed and replaced by a 2 man crew.

3.4.11 Brakes. Brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.42(b), 393.43 and 393.45 through 393.52. FMVSS 121 does not apply. Brake hose and brake hose assemblies shall conform to FMVSS 106. Braking requirements shall be met with pusher axle in the up or the down position.

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3.4.11.1 Service brakes. Vehicle shall be equipped with full air brakes. The braking system shall be complete with all necessary components and include:

a. Flange mounted air compressor, unloader-head-type, engine driven and engine lubricated air or water cooled, and having a capacity of not less than 12 cubic feet per minute (cfm) and 120 PSI at governor shut-off and 150 PSI relief valve opening.

b. Air storage reservoir(s) with not less than eight times the actuator displacement volume at maximum travel of the piston or diaphragm total capacity. Each tank shall be equipped with drain and safety and check valves between compressor and last reservoir tank.

c. Foot control suspended or treadle type.

d. Air pressure gage, visible to the driver.

e. Low air pressure warning buzzer.

f. Alcohol aspirator with unbreakable transparent container.

g. The wet tank shall be equipped with a drain valve that is readily accessible from the side of the vehicle.

h. Air control valves.

3.4.11.2 Trailer brake control system. A trailer brake control system shall be furnished that consists of the following:

a. Identification of emergency and service lines.

b. Coincident control of trailer brakes with prime mover foot control.

c. Independent hand control for trailer brakes.

d. Prime mover protection valve with dash control and automatic breakaway feature.

e. Trailer stoplight operative with foot brake & with hand control for trailer brakes.

f. Two connecting air hoses at least 110 inches in length, equipped with coiled spring hose guards and "glad hand" quick connector on trailer end of hoses. Hoses shall reach the end of the vehicle frame.

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3.4.12 Cab. Manufacturer shall furnish a standard conventional commercial model cab (with rear window) with a minimum of 112 inches BBC (bumper to back of cab). A 3 man crew cab shall be provided. Cab shall have an upholstered, individually adjustable driver's seat and a passenger(s) seat of at least 32 inch width. The individual driver's seat shall be of the Bostrum Viking T-bar type completely adjustable both fore and aft and to the driver's weight. Grips shall be provided on each side of cab to assist personnel to enter or leave cab and to assist personnel in climbing onto the truck tractor deck plate. Interior lighting shall be provided. Fastenings and seat belts conforming to FMVSS 208, 209 and 210 shall be installed for each seated crew member. When step height exceeds 24 inches, a secondary step shall be provided. Cab fenders shall have non-skid safety deck plate installed to permit walking on the fenders during vehicle servicing unless a tilt hood system is provided. Both doors will be provided with a suitable key lock. Two keys for each vehicle will be provided. The doors and ignition will utilize the same key and be interchangeable with all other XM911 vehicles.

3.4.12.1 Vehicle cab interior. The vehicle cab interior and upholstery color shall be black, dark green or dark brown; if these colors are not available then vehicle may be equipped with manufacturer's standard color. All vehicles produced shall have the same interior color(s). Interior plastic and metal surfaces shall conform to manufacturer's standard commercial practice. The fire wall shall be insulated to reduce engine noise and heat. Sunvisors for both driver and passengers shall be provided.

3.4.13 Steering. Power steering shall be furnished and have dry park capability at rated GVW. A mechanical connection between steering wheel and front axle steering mechanism must be in effect under all conditions such that if there is a power steering failure the vehicle can be manually steered and brought to a safe stop. Power steering must have 25 micron (or less) filter for this system.

3.4.14 Windshield wipers and washers. Vehicle shall be equipped with dual windshield wipers and windshield washers. Windshield wipers and eashers shall be multi-speed type and operated by air. Washers reservoir with water and appropriate additives for the climatic conditions encountered shall be furnished. Windshield wipers and washers shall conform to FMVSS 104.

3.4.15 Bumpers. Unless bumper is an integral part of vehicle cab, a channel type front bumper shall be provided. A radiator grille guard shall be provided.

3.4.16 Stowage. Stowage space with locking device will be provided to accommodate cables listed in paragraph 3.4.2.4.2 and items listed in paragraphs 3.4.16.1 and 3.4.16.2 with the exception of fire extinguisher, and first aid kit which will be vehicle mounted.

3.4.16.1 Tire changing tools. Each vehicle shall be furnished with tools required for exchanging mounted tire assemblies.

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3.4.16.2 Other equipment. Each vehicle will be furnished with the following equipment except for the last two items (items j & k) which will be furnished with the first two production vehicles only:

a. Hydraulic jack, 30 ton with handle for lifting the tractor or the M747 trailer to exchange mounted tire assembly with spare tire assembly.

b. Wheel-nut lug wrench, with handle for wheel removal and installation.

c. Utility chain, steel alloy, single leg, 3/4 inch diameter links, 14 feet long with one grab hook and 2 coupling links with sufficient strength for vehicle recovery.

d. Fire extinguisher, hand CF3BR, 2-3/4 pound capacity (vehicular mounted), rear of cab left side.

e. Tire pressure gage, self-contained (10 to 120 psi) suitable for checking tire pressures on tractor and M747 Semitrailer.

f. Pneumatic hose, tire inflator, 40 feet long with quick disconnect coupling and necessary fittings for inflating tractor and M747 trailer tires.

g. Warning device, highway, triangular, reflective, with container, Set 3, RR-W-1817.

h. Tools required by operator or crew for pre-vehicle check if applicable.

i. First aid kit, general purpose, 12 unit, in accordance with drawing 11677011.

j. Jumper cable, 20 feet long, electrical 28 volt for slave start from another vehicle.

k. Tire chains for the 4 outside dual wheels.

3.4.17 Cab heater and defroster. Manufacturer's standard heater and defroster with controls capable of providing adequate heat and window visibility in the temperatures to -25°F.

3.4.18 Controls and operating mechanisms. Manufacturer's standard controls and operating mechanisms shall be provided. Instruments, controls and mechanisms shall be identified as to their function.

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3.4.19 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard commercial equipment by the manufacturer. The following minimum equipment shall be furnished:

a. Ignition switch, key type using same key as cab door.

b. Voltmeter.

c. Fuel gage.

d. Engine oil pressure with warning light and buzzer.

e. Engine coolant temperature gage with warning light and buzzer.

f. Electronic speedometer with odometer.

g. Electronic tachometer.

h. Ash receptacle.

i. Air pressure gage with warning light and buzzer.

j. Standard commercial engine shut down fuel valve.

k. Transmission temperature and oil pressure warning devices as recommended by transmission manufacturer.

l. Interior light switch.

m. Seven day recording electronic tachograph that records engine RPM's and MPH's (may be combined with f and g above).

3.4.20 Rear view mirrors. Large full length (6" x 16") standard heavy duty west coast type outside rearview mirrors, each with at least three supporting arms shall be mounted on each side of cab. Mirrors shall be positioned such that driver may have clear view past the sides of an M60A1 tank (see MS-53064 for M60A1 characteristics) when being transported on the M747 Semitrailer. Combination (flat and convex) mirrors, inclosed in a common housing, shall be furnished. Each combination mirror shall have at least 50 square inches of flat reflective area and a separately adjustable convex surface having at least 20 square inches of reflective area. Each mirror assembly shall be capable of folding flat against the side of cab.

3.4.21 Fifth wheel. Vehicle shall be equipped with a full oscillating, 36 inch diameter fifth wheel with forks and semi-automatic lock for SAE J848 kingpin. Built-in lock-out for locking side oscillation shall be furnished.

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The fifth wheel shall be capable of being uncoupled by the operator while at the side of the 5th wheel. Uncoupling action shall be protected by a secondary manual lock, preventing movement of the uncoupling lever until the secondary lock is manually released.

3.4.21.1 Fifth wheel location. The fifth wheel shall be located to provide clearances specified under paragraph 3.2.9. Centerline of kingpin shall be such that the front axle will have 18,000 to 20,000 lbs at the wheels when loaded to actual GVW with the pusher axle in the up position.

3.4.21.2 Mounting. Fifth wheel mounting shall conform to Motor Carrier Safety Regulation 393.70(a), (c) and (g).

3.4.21.3 Approach ramps. Heavy duty approach ramps or plates of high tensile alloy steel shall be furnished to give support for fifth wheel forks and a continuous incline for semitrailer approach. The ramps or plates shall extend from the rear of the chassis frame to the fifth wheel forks. Approach ramps shall be heavy enough to support and be compatible with the M747 Semitrailer with a 60 ton load. The approach ramp shall be in a V shape that will guide the kingpin to the fifth wheel forks. The rear opening of the V shall not exceed 18" to insure that the narrow gooseneck on the M747 trailer can properly ride up the ramp. Ramps shall be of the box section type to preclude damage to the M747 Semitrailer kingpin. Stops shall be provided for the 5th wheel that preclude damage to the vehicle.

3.4.21.4 Deck plate and fenders. A self-cleaning grating of sufficient structural strength (for use while connecting air and electric lines between truck tractor and semitrailer and while operating the winches) shall be installed. Grating shall extend across and shall be secured to the frame rails with sufficient overhang to cover and protect side mounted tanks and fittings. Grating shall be located as close to the cab as possible and shall extend towards the rear of the vehicle. Provisions to allow access of personnel climbing onto the deck plate shall be furnished. Access for maintenance of fittings and other equipment shall be furnished. Fenders supported by solid support brackets shall be provided over the rear wheels to prevent road splash to tractor, top side surfaces and components. Fender and decking shall be capable of being walked on and will not interfere with wheel removal and installation. Fenders may terminate near the centerline of the front driving bogie axle if required to meet the requirements of 3.2.9(a).

3.4.21.5 Hose tender and cable supports. A pogo stick type hose tender with dummy gladhand connectors (to retain hoses when not in use) shall be provided behind the cab to accommodate and secure the trailer air hoses and electrical cables. The pogo stick type hose tender shall be mounted on the truck tractor deck plate or chassis members. Cable supports and stowage in accordance with best commercial practice shall be provided.

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3.4.22 Winches. The vehicle will be provided with two 45,000 lb. (bare drum) capacity hydraulic winches. The winches shall be hydraulically driven without the use of chains or other mechanical devices. The winches shall have two speeds and use either a worm or planetary gear reduction system. The low speed shall be not less than 13 feet per minute at full load on the first layer of the winch drum. The no load high speed must be at least twice as fast as the low speed. The winches with appropriate wire rope shall be mounted on the vehicle in such a manner that both winches may be used simultaneously (up to 45,000 pound total load) to load and unload the trailer. The wire rope for each winch shall be 150 feet in continuous length without splice joint. End of cables shall be equipped with clevis ends. All exposed moving, reciprocating and rotating power transmission devices shall be guarded. Winches must be mounted on the vehicle in such a way that they can withstand full capacity pull without damage to the vehicle frame or components.

3.4.22.1 Load line winches. At a winch drum pull rated capacity with a single line on bare drum the cable shall have a safety factor of not less than 2.0. Cable shall not be less than 1 inch in diameter. Power takeoff shall be of a capacity to operate winches simultaneously. The winch shall be protected with a hydraulic relief valve, set at 110% of rated winch load. Power takeoff shall have a horsepower rating of at least 90 hp for the hydraulic pump when operating off the converter. Both winches must operate simultaneously in high speed mode and the hydraulic system power requirement must have a minimum 90 hp capability without exceeding the PTO capacity.

3.4.22.2 Winch compatibility. The winches shall be mounted outboard on the truck tractor to provide compatibility with the M747 Semitrailer cable roller guides.

3.4.22.3 Winch operator's station. One winch operator's station with engine throttle control (or an automatic control that maintains engine speed under load) and controls adequate for safe and separate operation of both winches while operator is standing shall be provided. The winch operator's station shall be positioned between the vehicle cab and the winches. All controls shall be of the dead man type that will revert to control. Lever controls shall be speced at least 4 inches apart. Level control handles may be "doglegged" to satisfy this requirement.

3.4.22.4 Hydraulic reservoir. Vehicle shall be provided with a hydraulic reservoir, of capacity as recommended by the winch manufacturer and capable of withstanding the shock and vibration of off highway travel. Reservoir shall be provided with at least the following:

a. Filter(s), readily acceptable for cleaning or replacement without draining the reservoir.

b. Baffles to preclude foaming.

c. Dip stick and vented filler cap.

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d. Cleanout of a size to manually clean reservoir.

e. Hydraulic fluid to meet climatic operating requirements.

3.4.23 Drain plugs. Drain plugs installed in engine, transmission, transfer case, and axles shall be of the permanent magnet type.

3.4.24 Controls and control cables. All control cables going outside the cab shall be of the low friction type protected at both ends with adequate seals to prevent entry of moisture and contamination into the support tube and to provide a bearing surface for smooth motion of the end rod. The shift control tower shall be so constructed as to insure positive identification of the gear/range positions; shall have adequate devices to insure that shifting cannot take place accidently; and be illuminated in such a manner that the various gear/range positions can be seen at night while driving.

3.4.25 Wheel splash and stone throw protection. Removable rubber mud flaps shall be installed to the rear of the rear wheels in accordance with normal commercial practice. Mudflap installations at rear wheels, shall conform to the rear wheel splash and stone throw protection provisions of SAE J682.

3.5 Requirements, Special.

3.5.1 Initial production vehicle(s). The first two vehicles produced will be used by TARCOM in its publication/training program. Three of the next nine vehicles shall be selected by the Government for Initial Production Evaluation and designated as "Initial Production Vehicles" (see 6.3). Vehicles selected for Initial Production Evaluation shall be fully representative of the vehicles proposed to be furnished under contract. These vehicles shall be submitted for inspection to determine conformance to this specification. If submitted vehicles meet all requirements of this specification, no modifications shall be applied to subsequent vehicles to be produced under the production contract without prior approval by the Government except for product improvements.

3.6 Servicing and adjusting. Prior to acceptance of the vehicles by the Government, contractor shall service and adjust each vehicle and its mounted equipment including at least the following: winch system; focusing of lights; adjustment of engine; adjustment of electrical and brake systems; alignment of front wheels; inflation of all tires; complete lubrication of chassis, engine, running gear, and mounted equipment with grades of lubricants recommended for the ambient air temperature at the delivery point; servicing of windshield wiper reservoir with water and appropriate additives. A minimum of 25 gallons of fuel shall be provided in the fuel tank(s).

3.6.1 Engine coolant. The engine coolant shall be serviced with a solution of ethylene glycol and water in equal parts (to meet -25°F) by volume and an additive to prevent electrochemical type action in the cooling system.

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3.7 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Welded, bolted, and riveted construction utilized shall be in accordance with the highest standards of the industry. All components used in the assembly of the vehicle shall be new.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor's inspection or quality control system. The contractor shall provide and maintain an effective inspection or quality assurance system acceptable to the Government covering the supplies under the contract. A current, written description of the system shall be submitted to the contracting officer prior to initiation of production. The contractor shall notify the Government of any subsequent change to the submitted procedures prior to implementing the change into the system.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that practices, methods and procedures of the written inspection plan are being properly applied, and (b) Government inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures or instances of practices which might have an adverse effect upon the quality of the product will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct defects discovered shall be cause for suspension of acceptance until corrective action has been taken, or until conformance of product to specified criteria has been demonstrated.

4.2 Provisional inspection. Those vehicles selected by the Government for Initial Production Evaluation shall, prior to movement to a Government site, be subject to inspection at the Contractor's plant by representatives of the Government. The Contractor shall provide inspection assistance as may be required. At the time of this provisional inspection, the Contractor shall make available his inspection plan, inspection records and certifications pertinent to the vehicles and components.

4.2.1 Vehicle weight. One of the Initial Production Vehicles shall be weighed to determine curb weight and distribution of curb weight on front and rear axles with pusher in up and down positions. The imposed loading on front

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and rear axles will be computed and measured using the curb weight and payload prescribed herein to provide the specified GVW. Calculated and measured loads on front and rear axle will be utilized to ascertain that the suspension, axles and tires furnished are of adequate capacity to meet contract requirements.

4.2.2 Failure. Failure of the Initial Production Vehicles to meet requirements of the contract during the provisional inspection shall be cause for the Government to refuse acceptance of all vehicles under contract until corrective action has been taken.

4.2.3 Production sample. One of the first ten vehicles accepted by the Government shall remain at the manufacturing facility as a production sample, and be the last vehicle shipped on the contract. The Contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.3 Initial Production Evaluation. To determine conformance to Section 3 (inclusive) three vehicles will be selected by the Government for Initial Production Evaluation. This Evaluation is to be performed by the Government at a Government selected site and will last no longer than 60 days commencing with delivery of the vehicles to the evaluation site. The Contractor, at no additional cost to the Government, shall support the evaluation by prepositioning, at the evaluation site, minimum essential repair parts to include all anticipated wear-out and service parts to support the evaluation. Delays caused by vehicle break-down due to workmanship or material, failure of the Contractor to adequately support the vehicle with spare parts, or failure of the Contractor to comply with specification requirements, shall not be basis for adjustment of the contract delivery schedule or the contract price. All delays regardless of cause will extend said 60 day evaluation period by an amount of time equal to that needed to effect correction of failures and return said vehicle to a full evaluation readiness status.

4.3.1 Evaluation failure. Failure of the vehicle to comply with all requirements specified herein and all other requirements of the contract as a result of the initial production evaluation described above, shall be cause for rejection of the vehicle. Further, the Government may refuse to continue acceptance of production vehicles until evidence has been provided by the Contractor that corrective action has been taken to eliminate such defects. Such defect found during, or as a result of this evaluation shall be prima facie evidence that all vehicles already accepted prior to completion of the evaluation are similarly defective unless evidence satisfactory to the contracting officer is furnished by the Contractor that they are not similarly defective. Such defects on all vehicles shall be corrected by the Contractor at no cost to the Government.

4.3.2 Conformance verification.

4.3.2.1 Fuel, air, oil lines, hydraulic lines and fitting check. Air lines, fuel, hydraulic lines and fittings, shall be inspected for leakage at the place of maufacture. Installation shall be checked to insure that fittings are properly assembled and that no chafing with adjacent parts of the vehicle is evident.

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4.3.2.2 Winch and wire rope check. To determine conformance to 3.4.22 and 3.4.22.1, winch and wire rope shall be tested with load and checked for proper assembly, installation and functional requirements. Rated load of the winch system shall be checked at Government Proving Ground (GPG).

4.3.2.3 Environmental operation conditions tests. To determine conformance to high and low temperature operation, the vehicle shall be tested at a Government site to determine conformance to paragraphs 3.1.1.4, 3.1.1.5 and 3.1.1.5.1. To determine conformance to 3.4.1.1, the cooling system shall be checked to assure the system meets cooling requirements under fully loaded conditions. The maximum values of transmission oil temperature, engine coolant temperature and engine oil pressure, shall also be recorded during high ambient temperature and instrument tests conducted at GPG.

4.3.2.4 Road speed. To determine conformance to 3.3.1 and 3.3.2, the vehicle shall meet the specified requirements.

4.3.2.5 Grade and slope operation. To determine conformance to 3.3.1 and 3.3.5, the vehicle shall be operated on 3% and 20% grades headed up and down and on 10% side slopes with each side of vehicle up slope. Tests will be accomplished at a Government test facility.

4.3.2.6 Braking ability.

4.3.2.6.1 Service brake test. To determine conformance to 3.3.4, the brakes shall be checked for functional operation, stopping distance, grade holding ability as specified.

4.3.2.6.2 Parking brake. To determine conformance to 3.3.5, the brakes shall be checked for grade holding ability.

4.3.2.7 Electrical. To determine conformance to 3.4.2 through 3.4.2.6, with all lights and accessories functioning, the alternator voltage shall be checked at point specified and voltage shall be as specified. The lights and accessories and all auxiliary equipment shall be operated and checked or verified, for functional requirements.

4.3.2.8 Radio interference suppression test. To determine conformance to 3.4.2.6, the vehicle and components shall be subjected to the vehicle suppression requirements of specification specified.

4.3.2.9 Air/sound pollution level tests. To determine conformance to 3.2.2 the appropriate instrumentation shall be used to assure the vehicle meets specified requirements.

4.3.2.10 On-vehicle equipment. To determine conformance to 3.4.16, the OVE and stowage spece shall be checked for completeness of material and compartments.

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4.4 Comparison evaluation. The Government may select vehicles at any time during the production contract period and subject them to an evaluation to compare existing quality with previous standards. Provisions of paragraphs 4.3 and 4.3.1 apply, except that the evaluation period will not exceed 30 days and Contractor support is not required.

4.4.1 Comparison evaluation failure. Failure of any vehicle to comply with any of the requirements specified in the contract occuring during or as a result of the evaluation may be considered cause for refusal to continue acceptance of vehicles by the Government until objective evidence has been provided by the manufacturer that corrective action has been taken. Any defect found during or as a result of the evaluation shall be prima facie evidence that vehicles accepted subsequent to the previous acceptable comparison evaluation vehicle or initial production evaluation vehicle are similarly defective unless evidence satisfactory to the Contracting Officer is furnished by the Contractor that they are not similarly defective. Such defects on all vehicles shall be corrected by the Contractor at no cost to the Government.

4.5 Acceptance tests. To determine conformance of each vehicle to Section 3 and to the production sample (see 4.2.3) each Truck Tractor shall be inspected and operated by the Contractor for a distance of not less than 15 miles without payload. Contractor inspection and operation shall assure, as a minimum, proper assembly, adjustment and servicing and configuration, workmanship and operational conformity to the production sample with contractually authorized changes if any. The Contractor shall make available his inspection and service records applicable to each vehicle to the Government Inspector prior to final acceptance. The Government may inspect and/or operate vehicles prior to acceptance to verify Contractor tests.

4.5.1 Test failure. If a vehicle fails to pass the acceptance tests, the Government Inspector may stop acceptance of subsequent vehicles until evidence has been provided by the Contractor that corrective actions have been accomplished.

5. PREPARATION FOR DELIVERY

5.1 Vehicle processing. Vehicle shall be processed for shipment, from manufacturer's plant to initial receiving activity, in accordance with manufacturer's standard commercial practice.

6. NOTES

6.1 Intended use. The truck tractor vehicle is intended for use with the M747 or suitable substitute trailers. The primary mission of this transporter is to evacuate a disabled main battle tank or similar vehicles over primary or alternate main supply routings from vehicle collection points or maintenance establishments. This vehicle does not replace the tank recovery vehicle, but evacuates disabled vehicles after they have been put into position for pickup.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Arctic kit (3.1.1.4).

c. Inherent extreme environmental operation capability, if required (see 3.1.1.5.1).

d. Quantity of initial production vehicles (4.3).

6.3 Initial production vehicle. An initial production vehicle is one of the first ten(10) vehicles produced under manufacturing methods to be used in normal production as selected by the Government (see 4.3).

6.4 Performance prediction. Completed copies of SAE Truck Ability Prediction Procedure form and computation for each vehicle model furnished under contract should be submitted as specified in the contract. The SAE Work Sheet Item 1 should include vehicle model number, wheelbase, engine model number, and vehicle type. Unless other conditions are cited in the contract, computations should be made for normal atmospheric pressure, normal ambient air temperature, and still, dry air. The factors to be used in predicting truck ability (see 3.3.1) are established as follows for the corresponding SAE Truck Ability Prediction Procedure Tables:

Table 1	-	Tire Factor - This factor must relate to the size of tires furnished by the contractor in accordance with this specification.
Table 2	-	Altitude Factor - 1.00
Table 3	-	Rolling Factor - 1.398
Table 4	-	Area Factor - For truck tractor, use factor 0.272
Table 5	-	Velocity Factor - 182
Table 6	-	Altitude Factor - 1.00
Table 7	-	Chassis Friction Horsepower - Use applicable power unit GVW (at 60,000 plus interpolation between 24,000 and 26,000 to obtain 85,000 GVW) and the engine rpm (to nearest 100 revolutions) which is required for 45 mph speed. For engine speed not on this table, factors shall be extrapolated.

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Table 8 - Grade Factor - 0.83

Table 8A - Correction Factor - Not required

Table 9 - Road Factor - 0.2

NOTE: The above factors for Tables 3, 5 and 8 pertain to the 43 mph requirement on level road. For 14 mph on 3% grade, factors will be interpolated and selected from these Tables.

6.5 Definitions.

6.5.1 Standard commercial truck tractor. A standard commercial truck tractor is a model as depicted in the manufacturer's standard truck data book and which is comprised entirely of commercially proven components certified by the component manufacturer(s) and the vehicle manufacturer as being properly matched and rated to meet this specification. All chassis components shall be as represented in the chassis manufacturer's technical data and all body and mounted equipment shall be as represented in the body and equipment manufacturer's technical data. The technical data herein referred to shall be identical to the specifications and technical material furnished by the offerors to their authorized company representatives for use in the selection of vehicle models and components.

6.5.2 Commercially proven components. Commercially proven components are components which shall be of the latest model/series that have been manufactured, marketed and sold to the commercial public for at least one year prior to receipt of the first step of the procurement solicitation.

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Preparing activity:
Army - AT

Project No. 2320-A980

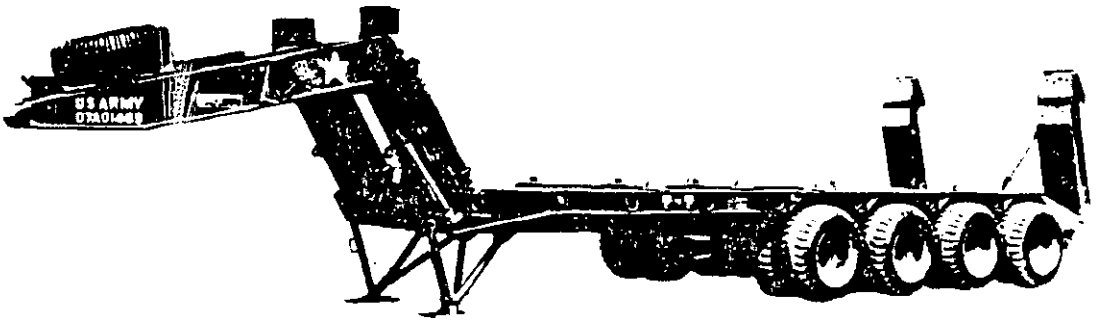
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CHARACTERISTIC DATA - SEMITRAILER — ARMY TANK-AUTOMOTIVE COMMAND

VEHICLE: SEMITRAILER, LOW BED, HEAVY EQUIPMENT TRANSPORTER, 60 TON, M747

STATUS: STANDARD A

REV. DATE: 2 DECEMBER 1975



FEDERAL STOCK NO. 2330-089-7265
PUBLICATIONS: . . . . . TM 9-2330-294-14
. . . . . TM9-2330-294-24P
PRIME MOVER: Truck, Tractor; 22 1/2 Ton, 8x8, XM746 or equivalent
FIFTH WHEEL HEIGHT: . . . . . 63 in.
KING PIN SIZE: . . . . . 3 1/2 in.
CURB WEIGHT (W/OVE) . . . . . 32,000 lbs.
WEIGHT DISTRIBUTION (W/60 Ton Payload):
5th Wheel . . . . . 46,000 lbs.
First Axle . . . . . 26,500 lbs.
Second Axle . . . . . 26,500 lbs.
Third Axle . . . . . 26,500 lbs.
Fourth Axle . . . . . 26,500 lbs.

SHIPPING DATA:
O.A. Length . . . . . 510 in.
Loading Platform Length . . . . . 317 in.
O.A. Width . . . . . 137 in.
reducible . . . . . 120 in.
O.A. Height . . . . . 107 in.
reducible-Dependent on loading
Square Feet 429
Cubic Feet-Dependent on loading
Tie Down Rings (Cargo) 12
Payload Anchors 8
Lift Points 4
BRIDGE CLASSIFICATION (XM746 Tractor & Payload): 76 "Caution Crossing"
INTERSECTION CAPABILITY (W/XM746):
. . 30 Ft. (Wall to Wall)
FORDING DEPTH: . . Dependent on Tractor
MIN. GROUND CLEARANCE . . . . . 9 in.
DEPARTURE ANGLE: . . . . . 27°
LOADING: . . . . . Rear End
RAMPS . . . . . Two Aluminum, Attached, Field Removable
DECK HEIGHT . . . . . 44 in.
TRACK GUIDES: . . . . . Eight
W/Twelve Locations and six positions

SUSPENSION:
First Two Axles - Solid Walking Beam
Last Two Axles - Trailing Arm w/Air *Springs & Air Lift*
AXLE TYPE: Rockwell 7 in. OD Tube, 1 in. Wall,

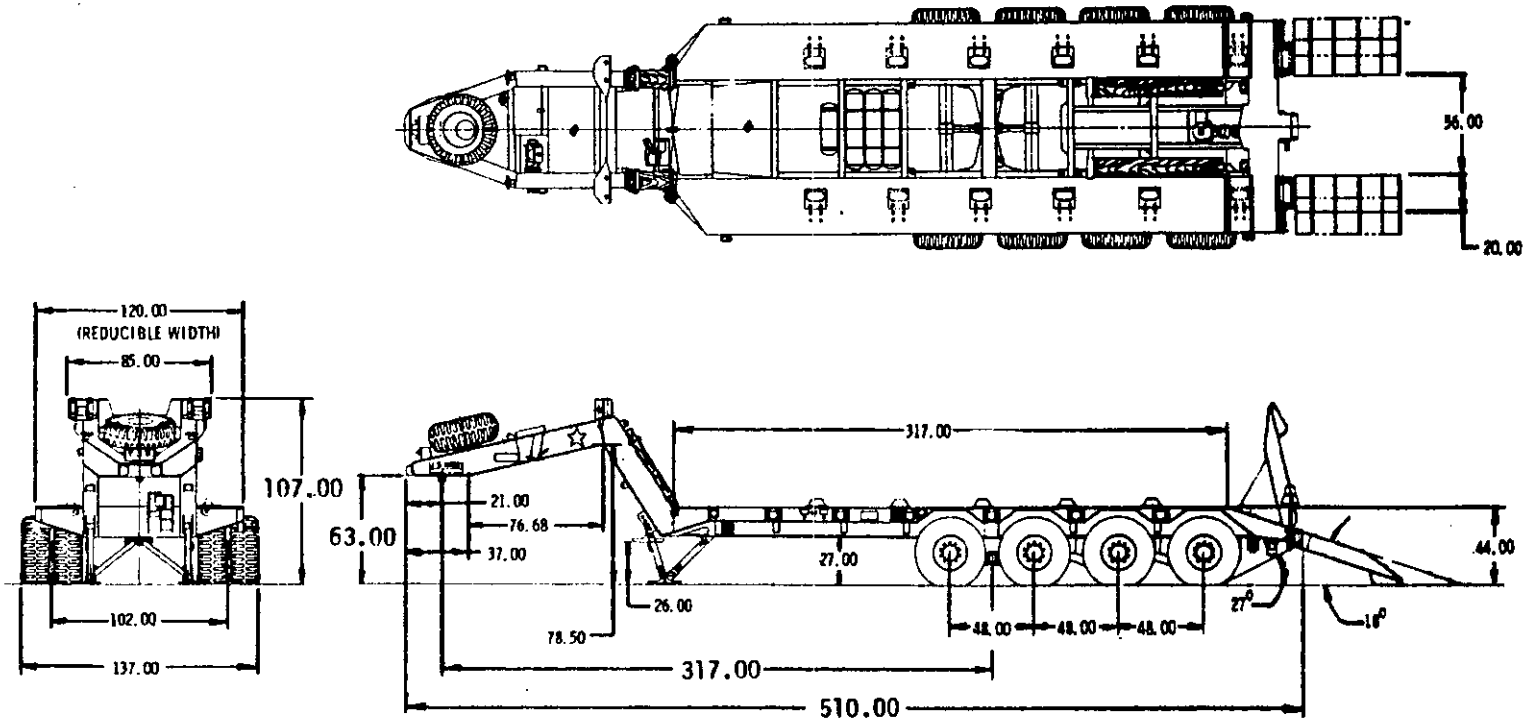
TREAD: . . . . . 102 in.
WHEELS: . . . . . Dual, 15° Drop Center, 11.75 x 19.5
TIRES:
Size 15-19.5
Type . . . . . Bias Ply, Tubeless
Ply . . . . . 14 PR
Tread Design . . . . . NDCC
Under Tread . . . . . Shreaded Wire
Recommended Pressure . . . . . 65 PSI
Spare . . . . . One
SERVICE BRAKES:
Number . . . . . Eight
Brake Size 15x7
Actuation . . . . . Air/vacuum (2 per wheel end)
PARKING BRAKES/FAILSAFES:
Number . . . . . Eight
Actuation . . . . . Spring
LANDING GEAR: Aluminum, Pin Type-Four Position plus stowage position
WINCHING PROVISIONS Four Cable roller assemblies and two snatch blocks provided.
ELECTRICAL SYSTEM: . . . . . 24 Volt
Includes service lights, blackout lights, and utility outlet.
STAKE POCKETS (Bulk Cargo) 16
WOOD DECKING KIT AVAILABLE

NOTES

PREPARED BY:	[signature] 5-23-72	TITLE: PROJECT ENGINEER
REVIEWED BY:	[signature]	TITLE: TRAILER BR CH.
APPROVED BY:	[signature]	TITLE: SYS DEV DIV CH.

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CHARACTERISTIC DRAWING



VEHICLE USAGE: The primary use of this Semitrailer (when used with suitable prime mover) is to evacuate a disabled main battle tank or similar disabled vehicles on axis of evacuation or from vehicle collecting points or maintenance establishments. This vehicle will not replace the tank recovery vehicle but will be utilized to evacuate disabled vehicles after they have been put into position for pickup. In addition, this vehicle will provide a means of transporting such vehicles from railhead, ports, depots, or supporting units to user maintenance areas. Secondary uses include transporting non-disabled tracked vehicles over extended distances when movement under their own power is not desired or feasible and transporting heavy engineer construction equipment and high density cargo, such as ammunition, gasoline, engines, and semi-mobile power plants of weight equivalent to the main battle tank.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-T-62274(AT)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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